

Thermophysical-property effects on melt-pool geometry and cooling histories in IN625 laser processing

Abstract

Geometric calibration makes thermal models useful for laser processing, but a fitted melt boundary leaves the underlying cooling history dependent on source and material assumptions. Here, a non-linear enthalpy conduction model is used to examine the thermal interpretation of length-calibrated bare-plate IN625 tracks. Linear continuation and holding at the last tabulated value are compared independently for conductivity and sensible heat capacity, with the calibrated source and phase law retained. Matching length leaves a wider and shallower fusion zone than the reference means. Within the controlled comparison, conductivity holding extends the pool, whereas heat-capacity holding shortens it. The conductivity-induced extension mainly displaces the rear solidus and liquidus together, with much less change in their separation. Across two scan speeds, nearly unchanged rear spatial phase spans correspond to a material passage time shorter by about one-third at the higher speed. These results distinguish translation of the thermal tail from enlargement of the phase region and connect spatial geometry to the time experienced by material. They provide a basis for interpreting property sensitivity after geometric calibration and for selecting thermal observations that assess the resulting histories.

Keywords: Laser powder bed fusion, IN625, Melt-pool geometry, Enthalpy model, High-temperature properties, Model discrepancy

1. Introduction

Melt-pool geometry is a practical link between laser processing conditions and the thermal models used to describe laser powder bed fusion (LPBF). The fused boundary constrains where melting occurred, while solidification analysis requires temperature gradients and the cooling histories experienced by material. A model that reproduces a track boundary can therefore support geometric process assessment. Spatially and temporally resolved thermography also provides observations of pulsed heating and cooling during solidification [1]. Geometric calibration supplies a boundary constraint, while the temperature and cooling behavior associated with that boundary remains dependent on source and material assumptions and requires independent assessment.

Moving-source heat transfer provides a useful foundation for addressing this question at manageable computational cost. The progression from concentrated-source solutions to distributed heating resolves the singular near-source behavior of classical solutions, while nonlinear conduction formulations incorporate temperature-dependent properties and latent heat [2]. Rapid transient calculations extend this approach to scan-path effects and have been compared with solidification conditions and grain-structure trends in selected laser and electron-beam powder-bed cases [3]. Their value lies in connecting process inputs to thermal conditions without resolving every melt-scale phenomenon. Fluid motion and deformation of the absorbing surface, however, redistribute energy through processes outside a conduction balance. Simplified models therefore rely on the heat-source representation and effective transport assumptions to reproduce the observed melt boundary [4].

Geometric calibration demonstrates both the usefulness and the coupling of these assumptions.

In the AMMT IN625 benchmark, imposing the measured laser profile reduced the freedom to adjust the source shape. A scalar absorbed-power factor then reproduced length more closely than width and depth; introducing calibrated directional conductivity brought the dimensions into closer agreement across the tested conditions [5]. This result shows that a well-constrained incident beam does not remove uncertainty in how heat is redistributed within a simplified model. An alternative approach represents unresolved laser–melt interactions through a volumetric source whose deposition depth and absorption respond to the calculated pool. For bare IN625 tracks, calibration across several beam diameters supported subsequent application to a layer-scale benchmark [4]. Together, these approaches establish that the calibrated geometry depends on the combination of source deposition and transport closure. Their fitted parameters consequently acquire meaning within that combination.

Temperature measurements provide a direct test of the thermal information supplied by such fits. For bare 316L tracks, several combinations of absorption and vaporization parameters in a fluid model gave comparable agreement with cross-sectional area and width-to-depth ratio, yet produced different surface-temperature profiles; two-color imaging distinguished among them [6]. The authors of the AMMT conduction study explicitly stated that their calibrated model was not valid for predicting the temperature distribution within the melt pool [5]. Geometric observations also constrain different features of a field. In the AMMT measurements, imaging length uses radiance-profile features at the front and rear, whereas width and depth come from metallographic cross-sections and separate observation populations [7]. Calibration to one of these quantities should thus be interpreted in terms of its measurement definition and its sensitivity to the modeled field.

Assessing a fitted field also requires distinguishing parameter response from observational agreement and physical feasibility. Analytical sensitivity calculations already quantify how laser inputs and constant thermophysical properties change pool dimensions and temperature histories [8]. These responses describe the adopted model; comparison with observations determines which outputs it can support. Combined thermography and metallography of 316L, IN625 and CoCr tracks found useful surface-geometry trends, while strong depth correlations coexisted with systematic deviations outside stable conduction melting [9]. This separates correlation from dimensional agreement and makes the operating regime part of the assessment. A complementary depth-based test inverts a specified conduction model for the absorptivity needed to reproduce measured penetration, then compares that value with its physical bound and tests alternative effective-transport assumptions [10]. The recovered factor serves as an attainability diagnostic under specified assumptions. Together, these studies distinguish sensitivity, output agreement and attainability, providing a basis for examining the constitutive assumptions that shape each modeled observable.

The material description supplies another basis for that interpretation. Conductivity controls diffusive heat redistribution, while sensible heat capacity enters the enthalpy stored during heating and released during cooling. These inputs are often available over only part of the temperature range reached by a laser track. For the IN625 supplier tables used here, conductivity ends at 982 °C and sensible specific heat at 1093 °C, below the listed 1290–1350 °C melting range. The bulletin labels conductivity as measured and specific heat as calculated typical alloy data [11]. A continuation is therefore required before either table can describe the phase-change interval or the hotter liquid. Broader assessments of nickel-superalloy properties also show that liquid-property estimates depend on the available measurements and estimation procedure [12]. A geometric fit cannot determine these constitutive functions independently of the source and transport model.

Continuing sensible properties must also be distinguished from specifying phase change. In an enthalpy formulation, latent heat contributes through the adopted phase-fraction relation as well as the total heat of fusion [2]. A linear fraction across a selected freezing interval is an established continuum approximation [4], but the interval itself depends on the material description. The supplier melting

range supplies one set of endpoints; an IN625 model intended for rapid additive-manufacturing cooling instead terminated solidification at a lower eutectic temperature derived from Gulliver–Scheil calculations, motivated by limited diffusion in the solid [11, 4]. These choices alter where latent storage is distributed along the temperature axis. They explain why a sensitivity to sensible heat capacity cannot be interpreted as an isolated latent-heat effect, and why comparisons of phase-region widths require a common phase law.

With that law fixed, a further distinction becomes available within the calculated temperature field. Total pool length combines the positions of the front and rear melting boundaries. At the rear, solidus and liquidus crossings describe both the location of the thermal tail and the distance over which material traverses the adopted phase interval. Moving both crossings downstream can lengthen the pool with little change in their separation. Changing their separation instead changes the spatial extent of that interval. Conductivity and storage influence the same coupled field, so a dimensional response alone does not identify which of these changes occurred. Resolving paired phase boundaries gives the geometric sensitivity a more specific thermal interpretation.

Converting that spatial structure to a cooling history requires the motion of material relative to the field. For a steadily translating temperature distribution, a stationary laboratory point traverses a rear spatial interval in a time equal to its length divided by scan speed. The constant-speed conversion is also used in the AMMT below-solidus cooling estimates [7]. It follows that similar spatial phase spans can correspond to different passage times and interval-mean cooling rates when speed changes. The steady mapping has a defined physical scope: pulsed LPBF measurements show time histories affected by successive heating events and post-pulse latent release [1], while fluid-tracer calculations show that melt motion changes the thermal paths followed by liquid parcels [13]. Material passage through a translating conduction field therefore provides a useful thermal descriptor whose interpretation depends on the frame and transport assumptions.

The present study examines these distinctions for three AMMT bare-plate IN625 conditions using a nonlinear moving-frame enthalpy model. One effective source factor is fitted to condition B surface length; width, depth and conditions A and C are compared using the same factor. A controlled comparison then replaces linear continuation by last-value holding independently for conductivity and sensible heat capacity, while keeping the source, phase law and numerical configuration fixed. The analysis separates the resulting length changes into front and rear boundary motion, resolves rear solidus–liquidus displacement and span, and relates the spatial span to material passage time. This design identifies how the chosen property continuations change the thermal meaning of a length-calibrated field. It connects geometric discrepancy, constitutive response and the derived cooling history to distinct observables that can guide further thermal assessment.

2. Materials and methods

2.1. Benchmark conditions and comparison design

The analysis connects surface-length calibration, fusion-zone shape and the rear phase interval through two complementary comparisons. A source factor fitted to condition B length is retained for all three operating conditions and for width and depth. At B, independent changes in the high-temperature conductivity and sensible-storage continuations resolve their contributions to the calibrated field. Rear solidus and liquidus positions then separate thermal-tail displacement from phase-span change, while scan speed converts the spatial interval to material passage time.

The benchmark comprises constant-speed single-track laser scans on bare IN625 plate under three public AMMT conditions (Table 1). Conditions B and C share the same nominal power and isolate a speed change; condition A combines a lower power with a lower speed. The measured beam diameter is $D_{4\sigma} = 170 \mu\text{m}$ [7]. The normalized circular Gaussian used here has $D_{4\sigma} = 2w$, giving $w = 85 \mu\text{m}$.

This reconstruction represents the measured second-moment diameter with circular symmetry, while the published measurement retains the two-dimensional beam map.

Table 1: Operating conditions and fixed-source energy scales. The effective factor $\eta \approx 0.28905$ is fitted to B length only. $E_\ell = \eta P/v$ and w/v are model input scales.

Condition	P (W)	v (m/s)	E_ℓ (J/m)	w/v (μ s)
A	137.9	0.4	99.65	212.50
B	179.2	0.8	64.75	106.25
C	179.2	1.2	43.17	70.83

Length references are the AMMT condition means reported by Lane et al. [7]. Width and depth references are the NIST challenge Table 2 means for the 100 μ s-track microscopy classes [14]. Lane’s later cross-section analysis combines the 100 and 20 μ s classes. Imaging length and metallographic cross sections consequently represent separately sampled condition-level populations. Their individual means provide the geometric comparisons used throughout the study.

The published geometric targets were consulted during model development. The calibration objective contains the condition B length; the resulting source factor is retained for B width/depth and for all condition A and C comparisons. Expanded uncertainties U with coverage factor $k = 2$ are taken from Lane’s uncertainty budgets. For width and depth, these scales belong to the published combined-class microscopy analysis, whereas the means used here belong to the NIST 100 μ s classes. Signed deviations δ are reported individually as δ/U relative to the corresponding published measurement scale. This reporting preserves the separate provenance of the mean and uncertainty and leaves model, input and calibration uncertainty outside the denominator.

2.2. Moving-frame enthalpy model and material continuations

The laboratory-frame conduction balance uses conservative enthalpy [2]:

$$\frac{\partial H}{\partial t} = \nabla \cdot [k(T)\nabla T], \quad H(T) = \rho \left[\int_{T_0}^T c_p(s) ds + \mathcal{L}f(T) \right]. \quad (1)$$

The density $\rho = 8440 \text{ kg m}^{-3}$ and melting limits $T_s = 1290 \text{ }^\circ\text{C}$ and $T_l = 1350 \text{ }^\circ\text{C}$ follow the IN625 bulletin [11]. The latent heat $\mathcal{L} = 280000 \text{ J kg}^{-1}$ follows the earlier AMMT model [5]; the reference temperature is set to $T_0 = 25 \text{ }^\circ\text{C}$. A linear liquid-fraction closure is adopted over that melting interval, following an established continuum-model approximation [4]

$$f(T) = \begin{cases} 0, & T < T_s, \\ (T - T_s)/(T_l - T_s), & T_s \leq T < T_l, \\ 1, & T \geq T_l. \end{cases} \quad (2)$$

Specific heat and conductivity are linearly interpolated between the input knots in Table A.4. The conductivity table contains 15 points and ends at 982 $^\circ\text{C}$; the 12-point specific-heat table ends at 1093 $^\circ\text{C}$. The supplier labels its typical specific heats as calculated and conductivity as measured [11]; these are general alloy data, not measurements on the benchmark coupon. Differentiating Eq. (1) within the adopted phase interval gives $c_{p,\text{eff}} = c_p + \mathcal{L}/(T_l - T_s)$, with a latent contribution of 4666.67 $\text{J kg}^{-1} \text{ K}^{-1}$.

Both property tables end below the adopted solidus, so the melting-range solution requires a high-temperature continuation (Fig. 1).

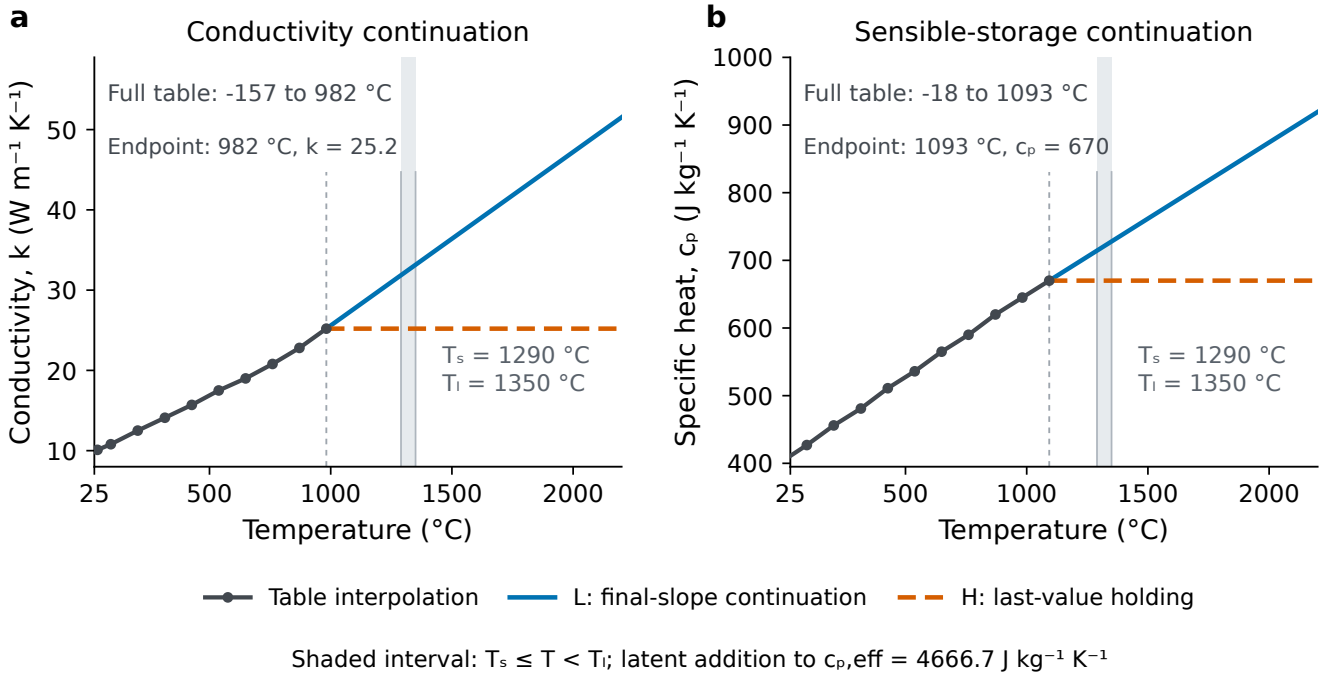


Figure 1: Actual IN625 property inputs and the two continuation choices. (a) Conductivity and (b) sensible specific heat use the input knots listed in Table A.4, with common piecewise-linear interpolation below the respective endpoints. The final knots are 982 °C and 25.2 W m⁻¹ K⁻¹ for k , and 1093 °C and 670 J kg⁻¹ K⁻¹ for c_p . L continues the final tabulated secant; H holds the final value. The displayed window is 25–2200 °C; all 15 conductivity and 12 specific-heat source rows, including those below 25 °C, are retained in the accompanying data and used in interpolation. Shading marks the adopted 1290–1350 °C phase interval. The latent addition to effective specific heat within this interval is 4666.7 J kg⁻¹ K⁻¹ and is stated separately from sensible c_p . The supplier describes conductivity as measured and specific heat as calculated typical alloy data [11]; the plotted high-temperature continuations are modeling choices.

Two binary factors replace linear continuation (L) by holding the last tabulated value (H), independently for k and c_p . H holds $k = 25.2$ W m⁻¹ K⁻¹ above 982 °C and $c_p = 670$ J kg⁻¹ K⁻¹ above 1093 °C. L extends the final tabulated secant of each property; the appendix lists the corresponding slopes. The changes apply above each table endpoint, including the solid range between the endpoint and T_s . Each conductivity branch changes k , its primitive K and the surface inverse together; each storage branch changes c_p , H and $T(H)$ together. The first letter labels conductivity. The baseline uses linear continuation for both properties (LL); HL and LH change one property, and HH changes both. The calibration and fixed-source contrasts are defined below.

Following the moving-source coordinate description [2], let $\xi = x - vt$. For a quasi-steady field, $\partial_t H = -v \partial_\xi H$; defining $\mathbf{u} = (-v, 0, 0)$ therefore gives

$$\nabla \cdot (\mathbf{u}H - k\nabla T) = 0. \quad (3)$$

The term $\mathbf{u}H$ represents stationary material passing through the laser coordinate frame. No liquid momentum equation or melt convection is included. The assumptions also exclude evaporation and free-surface deformation.

With the circular Gaussian $D_{4\sigma}$ convention [4], the surface input at $z = 0$ is

$$q(\xi, y) = \frac{2\eta P}{\pi w^2} \exp\left[-\frac{2(\xi^2 + y^2)}{w^2}\right]. \quad (4)$$

With outward normal $\mathbf{n}$, the top condition is $-k\nabla T \cdot \mathbf{n} = -q$; the Gaussian supplies inward heat. Its full-plane integral is ηP ; the appendix gives the discrete source integration. At positive ξ , incoming

material has $H = 0$; at negative ξ , diffusive gradient vanishes and enthalpy leaves advectively. The $y = 0$ plane is symmetric. Outer transverse and bottom boundaries are adiabatic. Radiation and ambient convection are omitted from the solved balance.

2.3. Melt-pool geometry and material passage

Model length L is the distance between two solidus crossings of the recovered surface symmetry-centerline temperature. Crossings are interpolated, must enclose the source, and must form one interval without domain truncation. Surface and symmetry-plane recovery precede threshold extraction; the appendix gives their numerical definitions.

Fusion-zone width and depth are extracted from the source-connected component of

$$T_{\max}(y, z) = \max_{\xi} T(\xi, y, z) \geq T_s. \quad (5)$$

Each ξ slice is reconstructed at the surface and symmetry plane, bilinearly interpolated onto observation coordinates, and then maximized over passage. This order defines the interpolated fusion envelope; maximizing nodal temperatures before interpolation gives a different envelope. Width is twice the positive- y extent and depth is the negative of the lowest z . The observation pitch controls sampling of the reconstructed geometry. Figures label the surface length proxy L_s and fusion-envelope extents W_f and D_f ; these correspond to L , W and D in the text.

Lane's imaging length uses a radiance-profile second-derivative minimum for the rear solidification location and a calibrated front threshold [7]. The model uses two solidus crossings of a recovered temperature field. The imaging procedure includes emittance, exposure and viewing effects through its radiance measurement; the temperature-threshold operator supplies the model boundary. The fitted η therefore belongs to the selected source, material law and boundary definition. Fusion-envelope width and depth are compared with metallographic measurements.

Let ξ_s^- and ξ_l^- denote the rear centerline solidus and liquidus crossings. Define the rear spatial phase span, passage time and interval-mean cooling-rate magnitude as

$$\ell_m = \xi_l^- - \xi_s^-, \quad \tau_m = \frac{\ell_m}{v}, \quad \bar{\dot{T}} = \frac{T_l - T_s}{\tau_m}. \quad (6)$$

These quantities characterize stationary material passing through the quasi-steady field. At a fixed laboratory point, $d\xi/dt = -v$, so the spatial interval and scan speed determine both the passage time and the interval-mean cooling magnitude. Constant-speed space-to-time mapping is also used in the AMMT measurements [7]; their cooling analysis uses a below-solidus interval, while the present calculation uses 1350–1290 °C. The three quantities in Eq. (6) share the same modeled temperature field.

2.4. Calibration, constitutive contrasts and numerical adequacy

Fitting $L_B(\eta) = 359 \mu\text{m}$ gives $\eta \approx 0.28905$, with a signed length residual of $+0.158 \mu\text{m}$. Conditions A and C retain this value, beam, phase law and property continuations. The appendix gives the scalar calibration algorithm, trial values and local sensitivity.

The four B corners share power, speed, beam, density, phase thresholds, latent heat, mesh, observation operators and the source factor of the LL calibration. The first letter denotes conductivity and the second sensible specific heat. Sharing B=LL between the operating-condition and continuation comparisons gives six production solutions.

For any output Q , define the baseline-anchored finite effects

$$\delta_k = Q_{\text{HL}} - Q_{\text{LL}}, \quad \delta_{c_p} = Q_{\text{LH}} - Q_{\text{LL}}, \quad (7)$$

$$I_{k,c_p} = Q_{\text{HH}} - Q_{\text{HL}} - Q_{\text{LH}} + Q_{\text{LL}}, \quad \Delta_{\text{joint}} = Q_{\text{HH}} - Q_{\text{LL}} = \delta_k + \delta_{c_p} + I_{k,c_p}. \quad (8)$$

The conditional conductivity effect at held c_p is $Q_{\text{HH}} - Q_{\text{LH}} = \delta_k + I_{k,c_p}$; the conditional storage effect at held k is $Q_{\text{HH}} - Q_{\text{HL}} = \delta_{c_p} + I_{k,c_p}$. Reporting both conditions establishes whether the response direction persists across the two factor levels. The effects are deterministic differences between prescribed continuation functions within one physical model.

An axial-face Péclet diagnostic compares coordinate enthalpy transport with diffusion in the phase interval. A discrete rear hot-region budget measures net outward diffusive power. Each branch defines its temperature-selected region separately, allowing the diagnostic population and region boundary to follow the solved field. These budgets complement the matched-coordinate phase thresholds; their definitions are given in the appendix.

The conservative finite-volume solution uses near-source cell widths of $4 \times 3 \times 1 \mu\text{m}$. The graded-domain geometry and enthalpy discretization are specified in the appendix. Convergence is checked using both the sum of absolute cell residuals and the signed global energy balance, followed by a full frozen-coefficient correction.

A separate constant-property moving-Gaussian problem checks source integration, frame transport and geometric extraction against a semi-infinite analytical reference. This verification isolates numerical implementation from assessment against physical measurements [15]. Its reference $L/W/D$ is $407.535/153.293/34.836 \mu\text{m}$; finite-volume differences are $+0.139\%$, $+0.190\%$ and -0.499% , respectively. The benchmark uses constant properties and omits phase change, making its accuracy assessment specific to that verification problem.

At the calibrated η , reducing the near-source axial spacing from 4 to $2 \mu\text{m}$ changes B length, width and depth by $+0.361$, -0.009 and $+0.041 \mu\text{m}$, and C by $+0.199$, -0.029 and $+0.045 \mu\text{m}$. A/B domain enlargement changes these quantities by less than $3 \times 10^{-6} \mu\text{m}$. These measured sensitivities support comparison with the larger geometric responses, but do not give a full three-dimensional error bound for each continuation branch. Submicrometre interactions are therefore reported descriptively. An earlier transient B comparison at a different calibration stage provides only approximate local steady consistency. Fixed-field blackbody estimates put B-corner top losses at 0.099 – 0.180% of absorbed power; feedback from these losses is not solved.

All six solutions satisfy the stated energy-balance and nonlinear-correction criteria and give bounded phase fractions and nontruncated extracted geometry. The appendix reports the acceptance thresholds and final residuals.

3. Results

The length-calibrated model is first compared with the remaining geometric measurements. The property contrast then separates changes in overall shape from motion of the rear phase boundaries, before the scan-speed comparison relates that spatial structure to material time.

3.1. Melt-pool geometry across scanning conditions

The length calibration gives $359.16 \mu\text{m}$ at B and leaves a systematic width/depth residual at B and C (Table 2 and Fig. 2). The B width exceeds the nominal reference by $19.54 \mu\text{m}$ and depth falls below it by $4.88 \mu\text{m}$, giving separate expanded-uncertainty-scale ratios of $+3.10$ and -1.23 . A closely matched axial length thus coexists with an over-wide, shallow fusion envelope.

Condition A deviations are smaller than their respective reported measurement-uncertainty scales. In C, the model is $49.37 \mu\text{m}$ shorter, $23.05 \mu\text{m}$ wider and $7.56 \mu\text{m}$ shallower than the nominal references. The predicted aspect ratios $W/D = 3.79, 4.60, 5.85$ for A/B/C also show the wide/shallow pattern, compared with condition-level reference ratios $3.48, 3.43, 3.58$. These reference ratios are calculated from separately sampled class means.

Table 2: Condition-level geometry comparisons in μm . U is the separately sourced Lane expanded uncertainty ($k = 2$). Reference width/depth means are current NIST 100 μs -track classes. B length is fitted. δ/U uses only the experimental scale, without combined model uncertainty.

Condition	Quantity	Model	Reference	U	δ/U
A	L	309.89	300	11.91	+0.83
	W	153.36	147.9	6.42	+0.85
	D	40.47	42.5	4.49	-0.45
B	L (fitted)	359.16	359	21.26	+0.01
	W	143.04	123.5	6.30	+3.10
	D	31.12	36	3.97	-1.23
C	L	320.63	370	32.57	-1.52
	W	129.05	106	4.48	+5.14
	D	22.04	29.6	3.26	-2.32

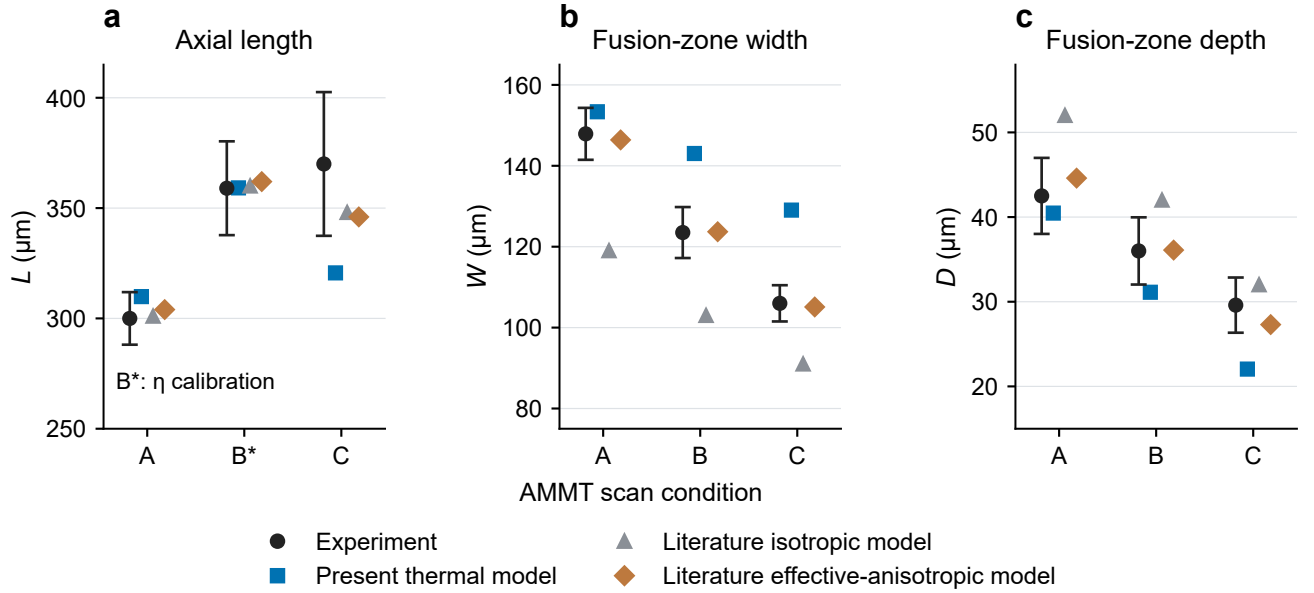


Figure 2: Geometry comparison with condition-level reference means and the published isotropic and effective-anisotropic calculations of Kollmannsberger et al. [5]. Experimental bars show separately sourced Lane $U(k = 2)$ scales; width/depth centers use current NIST means. B length is calibrated in the present model. Published-model points provide context: their beam representations, phase laws, calibration targets and observation definitions set the respective comparison scope.

At fixed power from B to C, a 50% speed increase reduces absorbed line energy by one third. Model width decreases by 9.78% and depth by 29.16%; the nominal reference changes are 14.17% and 17.78%. The modeled W/D increases by 27.36% from B to C. The discrepancy therefore involves both absolute dimensions and the balance between transverse contraction and penetration. Condition A also changes power, so the B–C pair provides the fixed-power speed comparison.

Nominal experimental length increases by 11 μm from B to C, whereas model length decreases by 38.53 μm . The two reported expanded-uncertainty scales give a zero-correlation quadrature of 38.89 μm , larger than the nominal experimental change. This descriptive comparison limits interpretation of the sign of that experimental trend; C’s individual discrepancy concerns its own reference mean. Retained axial changes below 0.37 μm are much smaller than the main B/C residuals,

indicating that further refinement in that direction would leave the principal shape discrepancy largely intact.

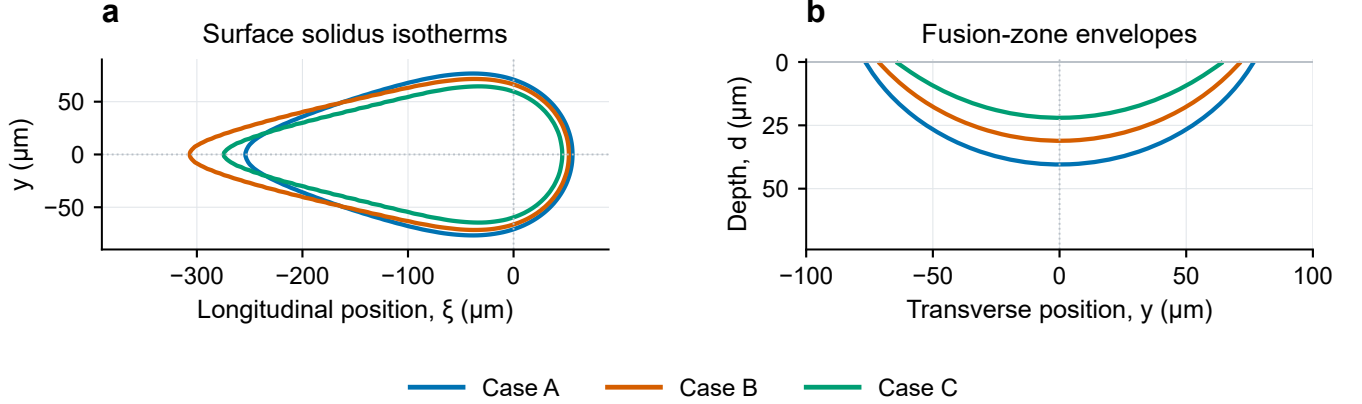


Figure 3: Model geometry for A, B and C. (a) Surface solidus contours in laser-frame coordinates. (b) Reconstructed maximum-temperature fusion envelopes in transverse/depth coordinates. Fusion envelopes use slice reconstruction and interpolation before passage maximization. Both panels display calculated thermal geometry and use the fixed calibrated source factor.

The contours in Fig. 3 show the preferential reduction in penetration for the fixed-power B–C comparison. The surface solidus contour determines the length proxy, while the transverse envelope records the maximum melted extent during material passage. Their differing responses motivate the property comparison at fixed source conditions.

3.2. High-temperature property effects on the thermal tail

The four-corner results are shown in Table 3 and Fig. 4. Holding conductivity at its last table value increases length from 359.16 to 402.37 μm at linearly continued c_p . Holding c_p instead decreases length to 352.99 μm . Holding both yields 393.44 μm . The joint increase is therefore

$$\Delta L_{\text{joint}} = 43.21 - 6.17 - 2.76 = 34.28 \mu\text{m}. \quad (9)$$

The heat-capacity response and interaction offset part of the conductivity response, revealing the competing contributions within the LL-to-HH difference. The signs persist at the other factor level: conductivity holding increases length by 40.45 μm when c_p is held, and storage holding decreases length by 8.93 μm when k is held. Conductivity holding therefore extends the tail and storage holding shortens it at both tested factor levels.

Conductivity holding moves the rear solidus and liquidus by -42.49 and $-43.50 \mu\text{m}$, while the front solidus moves by $+0.73 \mu\text{m}$. Rear solidus displacement accounts for 98.3% of the conductivity-induced length increase, with the front contributing 1.7%. The rear phase span changes by $-1.01 \mu\text{m}$. Heat-capacity holding moves the rear positions forward by $+6.22$ and $+6.53 \mu\text{m}$, changing their separation by $+0.31 \mu\text{m}$. The large length response predominantly translates the two rear phase boundaries together, while their separation remains comparatively stable.

At 1320 $^{\circ}\text{C}$, conductivity changes from 32.51 to 25.20 $\text{W m}^{-1} \text{K}^{-1}$ under holding, a reduction of 22.48%. Sensible specific heat decreases from 721.13 to 670 $\text{J kg}^{-1} \text{K}^{-1}$, or 7.09%. With latent storage included, effective specific heat decreases from 5387.79 to 5336.67 $\text{J kg}^{-1} \text{K}^{-1}$, or 0.95%, and enthalpy from the cold reference decreases by 0.66%. The property changes therefore perturb transport and storage by different amounts in the adopted mushy interval. The median axial-face

Table 3: Fixed-B deterministic continuation contrast. The first letter labels k , the second c_p ; L means linear continuation and H last-value holding above the respective endpoints. All branches retain the LL-fitted source factor. Rounded geometry and effects are in μm . Small transverse effects lack branch-specific refinement evidence.

Corner or contrast	L	W	D
LL	359.16	143.04	31.12
HL	402.37	143.15	30.09
LH	352.99	143.72	31.61
HH	393.44	143.79	30.80
δ_k	+43.21	+0.11	-1.03
δ_{c_p}	-6.17	+0.68	+0.49
I_{k,c_p}	-2.76	-0.04	+0.21
Δ_{joint}	+34.28	+0.75	-0.32

Péclet diagnostic rises from 4.44 in LL to 5.77 in HL, indicating a changed balance of diffusion and coordinate enthalpy transport on the selected faces. Integrated outward diffusion from the rear hot region also rises, from 14.724 to 14.913 W. Both the gradients and the temperature-selected integration boundary change with the solution, so this regional flux requires interpretation together with its moving spatial population. The conductivity branch also changes surface recovery through the corresponding material primitive.

The current-mesh depth contrast illustrates cancellation. Conductivity holding changes depth by $-1.03 \mu\text{m}$; heat-capacity holding contributes $+0.49 \mu\text{m}$ and the interaction $+0.21 \mu\text{m}$, leaving a net $-0.32 \mu\text{m}$. Conductivity effects remain negative at both storage levels (-1.03 and $-0.82 \mu\text{m}$), while storage effects remain positive at both conductivity levels ($+0.49$ and $+0.71 \mu\text{m}$). All corners retain the B width excess and depth deficit. The continuation choices thus change the tail substantially while preserving the principal fusion-shape discrepancy. The 34–43 μm length response exceeds the retained baseline axial sensitivity by a wide margin; the 0.04 μm width interaction remains an unresolved fine-scale comparison pending branch-specific refinement.

Across the four corners, the rear phase span lies between 79.25 and 81.17 μm while total length ranges from 352.99 to 402.37 μm . The contrast consequently changes the location of the rear phase interval much more than its spatial separation. This is the main comparison supplied by the retained centerline profiles; the small span ordering remains a current-mesh result.

3.3. Scan-speed dependence of the rear thermal interval

On the present mesh, the rear centerline phase spans are 48.30, 80.26 and 80.79 μm for A/B/C (Fig. 5). The B/C spans differ by only 0.53 μm , yet their transformed passage times are 100.33 and 67.33 μs . The C span is 0.66% larger, while the passage time is 32.90% shorter because speed increases from 0.8 to 1.2 m/s. The corresponding interval-mean cooling rate rises from 0.598 to 0.891 MK s^{-1} ; A gives 120.74 μs and 0.497 MK s^{-1} . Thus the B-to-C interval-mean cooling rate increases by 49.02% despite only a 0.53 μm difference in the spatial phase span.

The 49.02% increase follows from the shorter passage through an almost unchanged spatial phase interval, using Eq. (6). It therefore describes the thermal history along the modeled centerline; its dependence on scan speed cannot be inferred from the spatial span alone.

The fixed-power speed comparison distinguishes two responses of the modeled field: contraction of the fusion envelope and faster passage through the rear phase interval. Width and depth are compared with the condition-level metallography; the passage calculation characterizes the associated centerline temperature history.

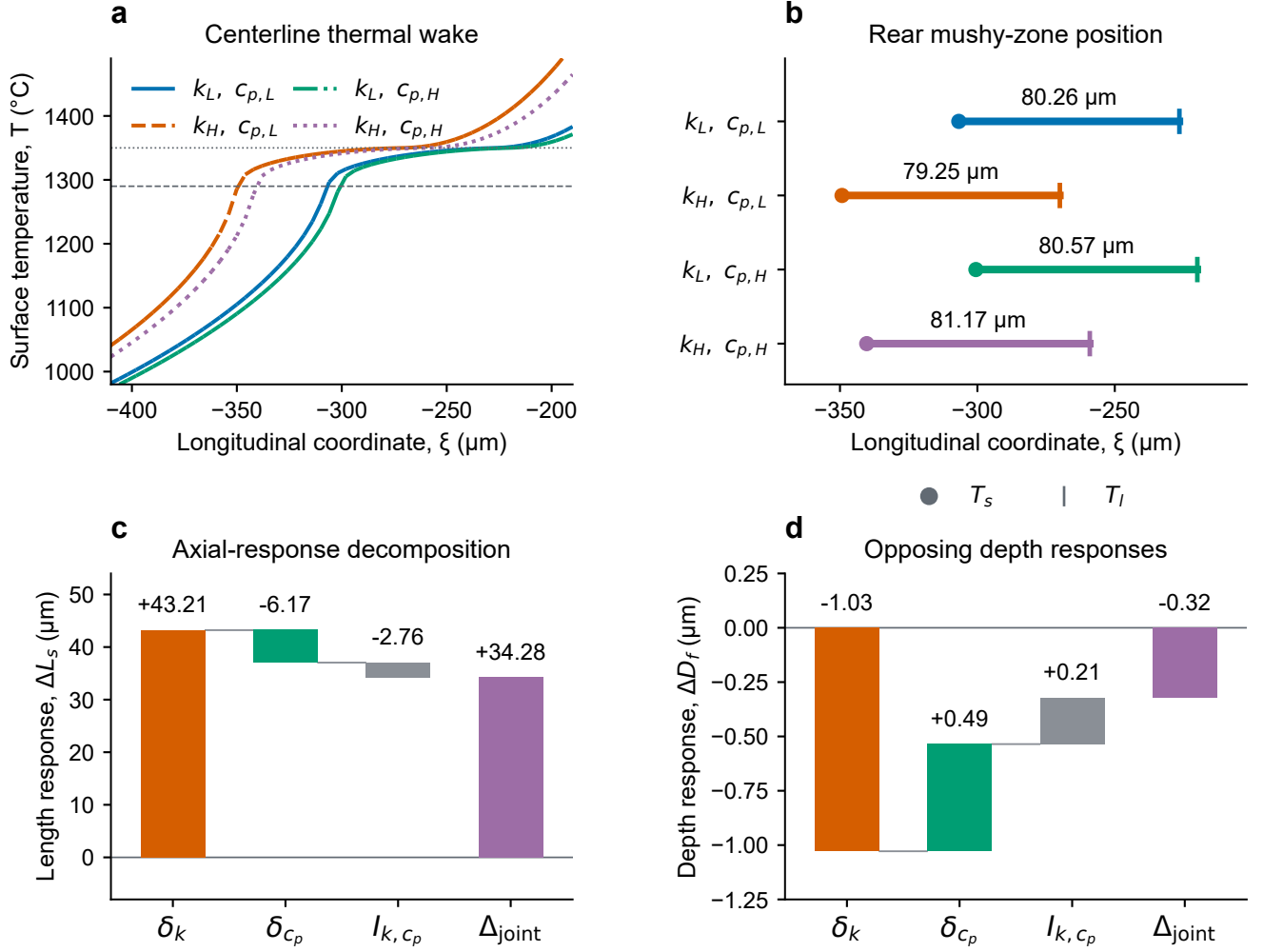


Figure 4: Four-corner continuation comparison at fixed B: rear centerline temperature profiles, phase-boundary positions and spatial span, and baseline-anchored geometry effects. The same phase thresholds, source factor and mesh are retained. Profiles are plotted from original field values with segment clipping to the displayed window. Factorial effects describe deterministic finite contrasts between continuation functions. Large rear translation dominates the length response; joint depth change conceals opposing single-factor contributions.

4. Discussion

4.1. Geometry calibration and model discrepancy

The length fit constrains a combination of source strength, material continuation and observation mapping. The retained B width excess and depth deficit, together with the B-C shape response, locate its remaining discrepancy in transverse heat distribution and penetration (Fig. 2). Surface length supplies one constraint on this field; fusion shape and temperature trajectories supply additional information. Source representation, transport closure and the relation between radiance-derived and temperature-threshold boundaries therefore enter the assessment as distinct contributors.

The information carried by geometry and temperature has also been separated experimentally. Bare-plate 316L calculations fitted several Fresnel/accommodation parameter combinations to cross-sectional area and width-to-depth ratio, producing different surface-temperature profiles; two-color measurements narrowed the acceptable combinations [6]. That study varied fitted parameter com-

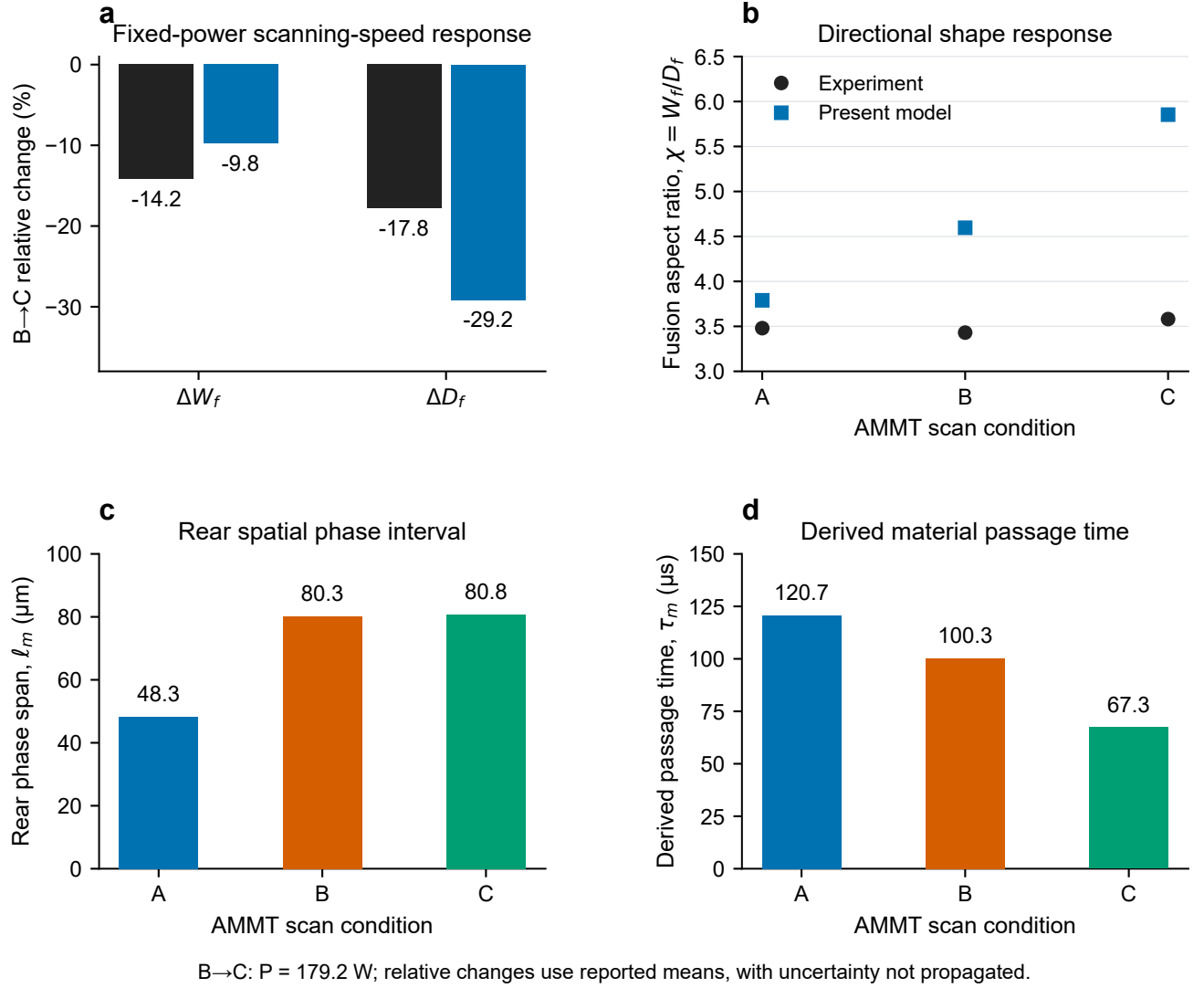


Figure 5: Operating-condition response. (a) Nominal B-to-C percentage changes in width and depth at fixed power, comparing model and condition-level reference means; uncertainty is not propagated. (b) Model and reference class-mean aspect ratios. (c) Model rear liquidus-to-solidus spatial spans. (d) Derived material passage times. Spatial spans and passage times are transformations of the same steady centerline field. The text reports the corresponding interval-mean cooling rates over 1350–1290 °C.

binations, whereas the present continuations retain the source factor of the LL length fit. Their contrast measures material-function response after a common calibration. Together these designs show the distinct roles of geometric fitting and temperature information in assessing a model.

The closest AMMT study achieved improved A/C geometry by combining measured laser input with effective directional conductivity calibrated at B [5]. Its authors treated internal melt-pool temperatures as a separate assessment problem. The measured source and directional closure provide a physically motivated comparison with the present isotropic Gaussian calculation. Retaining a common source and transport structure here isolates the response to the high-temperature material continuation.

4.2. Conductivity and storage in the thermal tail

Keeping the calibrated source fixed separates opposing property responses. Conductivity holding extends the pool at either storage setting, while sensible-storage holding shortens it at either conduc-

tivity setting (Table 3). Their combination gives a $34.28 \mu\text{m}$ length increase. The rear profiles locate the dominant response: the rear solidus accounts for 98.3% of the conductivity-induced extension, and the liquidus moves almost the same distance. These paired displacements shift the thermal tail relative to the source while leaving a comparatively stable phase separation (Fig. 4). The width excess and depth deficit persist across the four corners, separating this tail sensitivity from correction of the fusion shape.

The effective-heat-capacity representation separates latent and sensible storage [2]. In the adopted phase interval, latent heat contributes $4666.67 \text{ J kg}^{-1} \text{ K}^{-1}$ to effective specific heat. The 7.09% sensible- c_p reduction at 1320°C consequently becomes a 0.95% change in latent-inclusive effective specific heat and a 0.66% change in enthalpy from the cold reference, compared with the 22.48% conductivity reduction at the same temperature. This energy accounting explains the unequal local perturbation sizes applied to transport and storage. All branches retain the same latent heat; changing sensible heat capacity also changes pre-melting enthalpy and the temperature–enthalpy inverse because its continuation begins below the solidus.

Conductivity holding changes the local transport balance and the temperature gradients together. The selected axial-face Péclet diagnostic increases, and the integrated rear-region conductive power also increases (Section 3.2). The latter is evaluated over a temperature-selected boundary that moves with the solution, providing a counterexample to equating a smaller conductivity with lower integrated outward power from that region. Surface recovery changes consistently through $K(T)$ and its inverse. The resolved tail displacement thus combines the field response and the threshold-recovery operator. A matched-region flux comparison or separation of field and recovery contributions would distinguish their respective roles.

4.3. Spatial phase span and material cooling time

Rear-boundary location, phase separation and material time describe different aspects of the calibrated field. The B/C rear phase spans are 80.26 and $80.79 \mu\text{m}$, while passage times are 100.33 and $67.33 \mu\text{s}$ as scan speed rises from 0.8 to 1.2 m/s (Fig. 5). Nearly unchanged spatial separation therefore accompanies a much shorter material interval. The 49.02% increase in interval-mean cooling magnitude is another expression of the same threshold-and-speed relation in Eq. (6). Conversely, translating both rear boundaries together changes overall pool length while preserving nearly the same material passage interval at a fixed speed.

Spatial profiles and material temperature histories also have distinct measurement roles. Thermography has resolved centerline profiles together with fixed-location temperature histories through scan pulses and turn events [1]. The AMMT analysis converts position to time at a known constant speed, using a below-solidus temperature interval and its radiance-based boundary definition [7]. Aligning a measured trajectory with the present $1350\text{--}1290^\circ\text{C}$ calculation would require the same temperature interval and explicit treatment of scan position, exposure and observation mapping.

The passage transformation follows stationary material crossing the assumed steady field. Fluid motion adds another path through the thermal field: calculations for a Ni–20Cr dispersion-strengthened alloy produced narrower near-surface temperature peaks and hotter subsurface excursions than the corresponding unmodified conduction histories [13]. This distinction links thermal-field assessment to the trajectories used in microstructure models. A separate transient-conduction study assessed geometric and solidification-condition calculations for AlSi10Mg laser and IN718 electron-beam cases, and compared predicted grain morphology with experiments for the IN718 case [3]. It also showed interface velocities governed by hatch progression in overlapping multi-line melts. The present v -based conversion applies to a straight, quasi-steady track and characterizes its centerline passage interval; flow-dependent paths and grain/segregation predictions require their corresponding physical descriptions and measurements.

4.4. Thermal interpretation and discriminating evidence

Source distribution and effective transport remain competing explanations for the shape residual. A dynamic volumetric source calibrated on bare IN625 tracks at 195 W and 800 mm/s across several spot sizes was subsequently applied to a layer case [4]. The source adjusts its depth and effective absorption with the local melt-pool state, while its phase-dependent properties and 1410 K eutectic endpoint differ from the present interval. Matching the beam, phase assumptions, calibration target and geometry operators would make source redistribution and material closure directly comparable. Replaying the imaging operator would first quantify the effect of radiance-derived boundaries on the fitted source factor.

High-temperature material data and omitted liquid-state processes bound the physical use of these sensitivities. The supplier tables end at 982 °C for k and 1093 °C for c_p [11]; L/H extends them by two prescribed rules. Composition-based estimates include IN625 conductivity and identify limited underlying liquid-state data [12], providing candidates for a material-specific comparison. The present conduction balance also omits evaporation, melt flow and surface evolution. Its LL, HL, LH and HH positive- y surface-sample peaks are approximately 2697, 3507, 2756 and 3721 °C. The extreme held-conductivity temperatures make the absent liquid-state processes consequential; those peaks serve as sensitivity outputs of the assumed balance, while independent thermal measurements are needed to establish their physical credibility.

The numerical evidence distinguishes the scale of the main response from that of the fine interactions. The 34–43 μm length changes exceed the inspected baseline axial sensitivity; resolving submicrometre interactions would require branch-specific three-dimensional refinement. The analytical benchmark, energy balance and solution correction assess numerical execution. Physical comparison additionally depends on the calibration history, separately sampled length and cross-section populations, and source-specific measurement-uncertainty scales described in Section 2.1.

The comparisons identify complementary evidence for the next assessment. A defined source and observation mapping, tracks or profiles reserved before fitting, and an independently measured temperature trajectory would test the calibrated field together with its width and depth. Composition- and state-matched high-temperature data would constrain the material continuation. Each observation addresses a different part of the interpretation: fusion shape assesses energy redistribution, rear boundaries locate the phase interval, and a temperature history assesses its passage in material time. The present fixed-source contrasts supply this separation within the assumed conduction balance.

5. Conclusions

Geometric calibration is useful for low-cost thermal modeling of laser processing, while its physical assessment requires the observables carried by the fitted field to be distinguished. In the present IN625 enthalpy calculation, a close surface-length fit coexists with a wider and shallower fusion envelope than the reference means. The fixed-power speed comparison further exposes a discrepancy in the balance between transverse contraction and penetration. Fusion shape consequently provides an assessment of energy redistribution beyond the fitted axial extent.

Independent continuation contrasts reveal how high-temperature transport and storage assumptions contribute to that field. Conductivity holding extends the pool and sensible heat-capacity holding shortens it at both tested factor levels, giving a joint length increase of 34.28 μm . Rear solidus and liquidus translation dominates the conductivity response, while their separation changes comparatively little. The principal sensitivity therefore concerns the location of the thermal tail; the retained width/depth discrepancy calls for additional source, transport and observation evidence.

Material cooling time adds a further constraint to this spatial interpretation. Almost equal rear phase spans at 0.8 and 1.2 m/s correspond to a roughly one-third reduction in passage time and a

49.02% increase in the modeled interval-mean cooling rate. For each condition, the relation follows from passage through its calculated quasi-steady field and defines a derived centerline history. Taken together, the comparisons distinguish fusion shape, tail position and material time as complementary targets for thermal-model assessment. Credible high-temperature material data and a temperature trajectory aligned with the source, phase interval and observation definitions would test their physical correspondence.

Data and code availability

The LaTeX manuscript, vector figures, extracted tables and evidence inspection are available in the [manuscript package](#). The original case and numerical implementation are referenced through [SimAgent](#). Experimental sources remain the cited NIST challenge and Lane study. Publisher full-text copies used for reading are not redistributed. The package README distinguishes extracted tables and figures from large retained fields and identifies the corresponding original case locators.

Declaration of generative AI use in manuscript preparation

OpenAI Codex was used to assist literature and evidence organization, manuscript drafting and automated pre-review. Figures were generated from numerical results by deterministic plotting.

Appendix A. Material data and numerical verification

Appendix A.1. Material-property inputs

Table A.4: Thermophysical-property inputs from the general IN625 bulletin [11]. The two temperature columns describe different tables. Conductivity and sensible specific heat are interpolated independently.

T_k (°C)	k (W m ⁻¹ K ⁻¹)	T_c (°C)	c_p (J kg ⁻¹ K ⁻¹)
-157	7.2	-18	402
-129	7.5	21	410
-73	8.4	93	427
-18	9.2	204	456
21	9.8	316	481
38	10.1	427	511
93	10.8	538	536
204	12.5	649	565
316	14.1	760	590
427	15.7	871	620
538	17.5	982	645
649	19.0	1093	670
760	20.8		
871	22.8		
982	25.2		

The continuation slopes from the final tabulated pairs are 0.0216216 W m⁻¹ K⁻² for conductivity and 0.225225 J kg⁻¹ K⁻² for sensible specific heat. The enthalpy integral is evaluated piecewise analytically and inverted using the corresponding property continuation.

Appendix A.2. Spatial discretization and observation recovery

The domain is $\xi \in [-1, 0.35]$ mm, $y \in [0, 0.36]$ mm and $z \in [-0.30, 0]$ mm. A graded orthogonal mesh contains 902664 cells. Near-source cell widths are $4 \times 3 \times 1$ μm within $\xi \in [-0.48, 0.16]$ mm, $y \in [0, 0.12]$ mm and $z \in [-0.06, 0]$ mm. Outer widths increase by a factor of 1.22 up to 25 μm . The half-domain mesh has $199 \times 56 \times 81$ cells. The solved problem is steady; retained warm fields provide nonlinear initial guesses.

The Gaussian source is integrated rectangle-wise using error functions to supply each top-face power, which is deposited in the adjacent top control volume. Its full-plane integral is ηP and the transverse half-domain receives $\eta P/2$.

Surface recovery satisfies

$$K(T_{\text{surf}}) - K(T_{\text{cell}}) = q \Delta z/2, \quad K'(T) = k(T), \quad (\text{A.1})$$

and the symmetry-plane value uses the even quadratic extension $T(0) = T(y_1) + [T(y_1) - T(y_2)]y_1^2/(y_2^2 - y_1^2)$ from the first two positive- y cell centers. The recovered surface temperatures and symmetry extension are applied before the threshold crossings are extracted.

The observation pitch is 0.5 μm ; it samples the interpolated geometry rather than changing the heat-field resolution.

Appendix A.3. Scaled enthalpy diffusion and diagnostic definitions

The scaled unknown is $U = H/C_0$, with $C_0 = \rho(402 \text{ J kg}^{-1} \text{ K}^{-1}) = 3.39288 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$. On an internal face,

$$D_f = \frac{k_f}{C_0} \frac{\Delta T}{\Delta U}, \quad (\text{A.2})$$

where k_f is the harmonic face conductivity and D_f has units of $\text{m}^2 \text{s}^{-1}$. For $|\Delta U| \leq 10^{-8} \text{ K}$, the derivative limit is $\Delta T/\Delta U = C_0/[\rho(c_p + \mathcal{L}f')]$ at the first adjacent cell. This coefficient recovers conductive flux in the conservative equation for scaled enthalpy U . Coordinate enthalpy transport uses an exponentially fitted finite-volume convection flux. Material coefficients and the diffusion-transport stencil are updated with each nonlinear iterate; Anderson acceleration solves the fixed-point residual and preconditioned GMRES solves the inner linear system. Software versions and iterative controls are given in the accompanying numerical reproduction notes.

The retained face transport diagnostic is

$$|\text{Pe}_f| = \frac{|\mathbf{u} \cdot \mathbf{n}_f| d_f}{D_f}, \quad (\text{A.3})$$

where d_f is the adjacent cell-center separation. Unweighted medians are taken over internal axial faces adjacent to at least one cell with $T_s \leq T < T_l$. The diagnostic compares coordinate transport with diffusion in the discretized field; its selected face population can change between solved branches.

The discrete rear hot-region budget selects cells with $T \geq T_s$ and $\xi < 0$. Net outward diffusive power is the sum of signed conductive face powers on the boundary of this selected cell set; internal face contributions cancel. The region is selected separately in each field, so its boundary and sampled faces change with the continuation. This accounting diagnostic is interpreted alongside the matched-coordinate centerline thresholds rather than as a fixed-region causal contrast.

Appendix A.4. Scalar calibration and solution acceptance

A safeguarded secant solves $L_B(\eta) - 359 \mu\text{m} = 0$ in the initial bracket $[0.2, 0.4]$, with proposals restricted to the inner 90% of the current bracket and an accepted absolute length residual below 0.5 μm . It yields $\eta = 0.2890548519203547$, with a residual of +0.158 μm . The calibration record

below gives trial values and the local sensitivity conversion. A and C retain the final value, the same beam and phase law, and the same continuation rules.

Nonlinear acceptance requires the sum of absolute assembled cell residuals, normalized by absorbed half-power, to be below 2×10^{-5} , the signed global balance magnitude below the same scale, and a further frozen-coefficient solve to change temperature by less than 0.002 °C. Local residual and global balance are retained separately because opposing local errors can cancel in a global sum.

The six production solutions have finite temperatures, bounded phase fractions, nontruncated extracted geometry and the prescribed integrated source power. Normalized absolute residuals are 6.90×10^{-8} – 1.31×10^{-7} , signed global balance magnitudes are below 6.1×10^{-11} , and full-solve temperature corrections are below 2.7×10^{-5} °C.

Appendix A.5. Separate constant-property verification problem

The Gaussian verification uses $P = 179.2$ W, $\eta = 0.2$, $v = 0.8$ m/s and $w = 85$ μm , with $\rho = 8440$ kg m^{-3} , constant $c_p = 410.94444$ $\text{J kg}^{-1} \text{K}^{-1}$, constant $k = 9.870588$ $\text{W m}^{-1} \text{K}^{-1}$ and no latent heat. Thus $a_0 = k/(\rho c_p) = 2.845886 \times 10^{-6}$ m^2/s . Its finite-domain mesh contains 178992 cells ($113 \times 33 \times 48$), with near-source spacings of $8 \times 6 \times 2$ μm . Domain extents follow the production geometry, but the adiabatic transverse and bottom boundaries truncate the semi-infinite analytical referent.

A constant-property moving-Gaussian reference is constructed from the semi-infinite heat kernel, in the moving-source formulation [2]. With $Q = \eta P$, convolving the surface Gaussian with the Neumann kernel gives

$$T - T_0 = \frac{Q}{\rho c_p \pi \sqrt{\pi a_0}} \int_0^\infty \frac{\exp \left[-\frac{(\xi + v\tau)^2 + y^2}{w^2/2 + 4a_0\tau} - \frac{z^2}{4a_0\tau} \right]}{\sqrt{\tau}(w^2/2 + 4a_0\tau)} d\tau. \quad (\text{A.4})$$

The archived implementation removes the surface singularity by $\tau = (w/v)s^2$ and uses adaptive quadrature. Solidus geometry is extracted at 1290 °C, with uniform-core face-average source recovery for the numerical surface. Width/depth use the monotone constant-property maximum-temperature envelope; they do not replace the general production passage observer. The associated recipes and archived reports are identified in the numerical reproduction notes. The quoted differences combine discretization, reconstruction and finite-domain effects, and do not bound nonlinear phase-change or physical-model error.

Appendix A.6. Calibration record and retained sensitivities

The calibration bracket $\eta \in [0.2, 0.4]$ has retained endpoint lengths of 234.57 and 511.68 μm . Subsequent retained trial pairs $(\eta, L/\mu\text{m})$ are (0.2898051, 359.96) and (0.2853148, 354.22), followed by the final (0.2890548519, 359.16). They establish a sampled local bracket, not global identifiability. The local secant sensitivity is approximately 1276.79 μm per unit η . Dividing B's 21.26 μm expanded uncertainty by this slope gives a scale $\Delta\eta = 0.01665$; it is not a posterior interval and is not propagated into the geometry comparisons.

For uncertainty-scale reporting, $\delta/U = (Q_{\text{model}} - \bar{Q}_{\text{source}})/U_{\text{Lane}}$. The separate U values are not specimen standard deviations. Imaging and microscopy have different operators and aggregation, so the geometry triplet is not fitted to a joint likelihood. All finite contrasts use full-precision outputs before rounding for display.

Table A.5: Retained calibrated axial sensitivities in μm , calculated as fine minus baseline. Only the near-source axial spacing changes from 4 to 2 μm ; transverse and depth spacings remain 3 and 1 μm . These contrasts do not bound every discretization component.

Condition	ΔL	ΔW	ΔD
B	+0.361	−0.009	+0.041
C	+0.199	−0.029	+0.045

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